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1.0 Introduction

1.1 Problem Statement

Studying health among undergraduate students is critically important because this demographic represents a high risk population with mental health disorder prevalence rates approximately 26.3% that significantly exceed those of the general public (Leow et al., 2025; Sheldon et al., 2021). Affected students report higher levels of psychosocial distress, depression, and poorer self-rated health compared to food-secure peers (Azmi et al., 2022). On top of that, empirical Malaysian research shows food insecurity acts as a material and chronic psychosocial stressor, negatively impacting psychosocial and psychosomatic health (Syed Elias et al., 2023). While this research holds theoretical and practical value in highlighting food insecurity's toll on students' psychosocial, physical health, and well-being, it lacks direct practical application in Malaysian university contexts, primarily exploring correlations between food insecurity, socioeconomic status, and academic performance (Mohd Faidi et al., 2023).

1.2 Objectives

This study aims to empirically examine the impact of food insecurity on the health outcomes of undergraduate students at the International Islamic University Malaysia (IIUM), Gombak campus. Specifically, the objectives are:

1. To estimate the effect of food insecurity on students' mental and physical health outcomes, specifically depression, body mass index (BMI/obesity), and self-rated health.
2. To examine the role of subjective social status as a psychosocial mechanism linking food insecurity to health indicators, in line with the Stress Process Model (Willis, 2021).
3. To empirically assess relationship between food insecurity among undergraduate students with environmental shocks (i.e floods), household income per capita, financial aid (ie zakat and scholarship) as well as academic schedules

1.3 Significance of Study

While existing literature identifies university students as a high-risk group for economic instability and psychological distress (Loofbourrow & Scherr, 2023; Nikolaus et al., 2020), significant gaps remain regarding the contributors to food insecurity specifically among Malaysian undergraduates. This study bridges that gap by connecting food insecurity with subjective social status and lifestyle habits—variables often studied in isolation. The resulting data will be instrumental for stakeholders, including universities and zakat providers, to move beyond generic aid and towards targeted, high-impact policy interventions that secure both the nutritional needs and long-term academic success of students.

2.0 Literature Review

2.1 Concepts

The United States Department of Agriculture (USDA) and the Office of Disease Prevention and Health Promotion (ODPHP) define food insecurity as "a household-level economic and social condition of limited or uncertain access to adequate food" (Becerra & Becerra, 2020). Expanding on this, the Food and Agriculture Organization (FAO) of the

United Nations emphasizes the necessity of nutritious foods for normal growth and an active life (Anthem, 2025). Contextualizing this locally, the National Coordinating Committee on Food and Nutrition (NCCFN) of Malaysia adopts a comprehensive framework in the National Plan of Action for Nutrition III, defining food security as a state where all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food that meets their dietary needs and food preferences for an active and healthy life (NCCFN, 2016).Furthermore, other perspectives highlight that food insecurity also encompasses limited access to socially acceptable foods, reflecting not only quantity and quality but also the manner in which food is obtained (Sulaiman et al., 2021).

2.2 Theory

This study is based on the stress process model, which explains how social and economic conditions affect people's health by exposing them to different levels of stress (Willis, 2020). In this model, food insecurity is seen as a continuous source of stress that can harm both physical and mental health. People who face food insecurity often experience poorer health and a lower quality of life, with more hospital visits and higher medical costs (Hanmer et al., 2021). Willis (2021) stated that people from lower social and economic backgrounds are more likely to experience food insecurity and have fewer resources to cope with stress, leading to poorer health outcomes. In Malaysia, studies show that university students who experience food insecurity often report higher levels of stress, anxiety, and depression. These emotional pressures also affect their academic performance, supporting the idea that food insecurity acts as a stressor that influences both mental well-being and learning outcomes (Ahmad et al., 2022).

2.3 Dependent Variable (Health Outcomes)

2.3.1 Self rated health

Self-rated health (SRH) is a widely utilized and validated indicator of general health status, providing a comprehensive assessment of an individual's physical health. It is typically measured by a single-item question asking respondents to rate their overall health on a Likert scale ranging from "poor" to "excellent" (McLoughlin et al., 2021). This subjective measure is highly regarded for its ability to capture unobserved health dimensions that clinical assessments may miss, making it a robust predictor of future morbidity, functional decline, and mortality. In the Malaysian context, Sahril et al. (2023) demonstrated the reliability of SRH as a health metric, finding that self-reported poor health is strongly correlated with the presence of chronic non-communicable diseases and functional limitations. Consequently, SRH serves as a practical and effective proxy for evaluating the general health status of the study population, as further detailed in the methodology section. This finding is corroborated by recent Malaysian research, which indicates that food-insecure households and individuals consistently report lower self-rated health scores, attributed to the psychological stress and nutritional deficiencies associated with financial instability (Sahril et al., 2023; Nurul Izzati & Azdie, 2022).

2.3.2 Body mass index (BMI)

Research indicates a complex relationship between food insecurity and Body Mass Index (BMI), suggesting that traditional linear models may fail to capture the dual burden of

malnutrition. In the Malaysian context, Nurul Izzati and Azdie (2022) observed a significant prevalence of overweight and obesity among food-insecure undergraduates. This finding supports the "resource substitution" hypothesis, where financially constrained students replace nutrient-dense meals with cheaper, energy-dense processed foods to curb hunger leading to weight gain despite economic hardship.Conversely, other studies highlight the opposite extreme. Sulaiman et al. (2021) noted that food insecurity can manifest as insufficient caloric intake and meal skipping, resulting in chronic fatigue and low energy. Lemp et al. (2024) confirmed that such forced fasting and irregular eating patterns cause significant weight fluctuations, often driving BMI into the underweight range. Consequently, the literature establishes that food insecurity negatively impacts student health in two opposing directions: it can drive higher BMI through the overconsumption of cheap, obesogenic foods, or lower BMI through forced caloric deprivation (Loofbourrow & Scherr, 2023). This distinct "double burden" necessitates a methodological approach that distinguishes between underweight and overweight outcomes rather than treating BMI as a linear continuum

2.3.3 Depressive symptoms

Depressive symptoms constitute a primary focus in assessing mental health outcomes within this analysis. The existing literature indicates a substantial association between food insecurity and heightened psychological distress, anxiety, and depression. Individuals experiencing food insecurity have significantly higher risks of depression and anxiety than food-secure individuals, with the relative risk of mental illness from food insecurity estimated to be almost three times higher than that of losing a job (Fang et al., 2021). To systematically evaluate these outcomes, depressive symptoms are typically measured using validated psychometric instruments, most notably the Center for Epidemiologic Studies Depression Scale (CES-D). The questionnaire assesses key domains including depressed mood, feelings of guilt, and somatic symptoms such as sleep disturbance and loss of appetite. Even in Malaysia, the CES-D has been validated as a reliable screening tool for the local population, demonstrating strong internal consistency and concurrent validity in identifying depressive symptoms among adolescents and young adults (Singh et al., 2021). Furthermore, recent local studies reinforce the link between food insecurity and mental health; Ahmad et al. (2022) found that psychosocial factors, including stress and depressive symptoms, mediate the relationship between food security status and academic performance among Malaysian university students. This underscores the critical need to analyze how food insecurity specifically exacerbates mental health vulnerabilities in this population.

2.4 Independent Variable (Health Outcomes)

2.4.1 Parent’s Income

Parental income has been repeatedly linked to better physical and mental health outcomes among adolescents and young people. In this study, the respondent was asked to state their parents income (in MYR). Higher-income families are more able to provide

adequate nutrition, stable housing, and access to healthcare and educational resources, which cumulatively promote better physical and mental health (Quan et al., 2024). Meanwhile, low-income households and financial constraints are consistently associated with higher food insecurity in Malaysian populations and notes that low-income groups often have poorer dietary intake and nutrition-related health risks, indicating a pathway from low income to poorer diet and health (Sulaiman et al., 2021). Furthermore, socioeconomic status among students also shows that those from higher socioeconomic backgrounds have better academic and psychosocial profiles, which are closely tied to mental well-being and perceived health status, reinforcing the hypothesis that higher parental income predicts more favourable health outcomes among undergraduates (Ismail et al., 2020).

2.4.2 Food insecurity

Food insecurity is a central social determinant of health among university students and has been consistently linked to poorer self-rated health, greater psychological distress, and adverse nutritional outcomes. In Malaysia, a systematic review of food insecurity highlighted that tertiary students are one of the most affected groups, with food insecurity associated with inadequate dietary intake and a higher risk of nutrition-related health problems (Sulaiman et al., 2021). A study of Malaysian undergraduates similarly reported that students experiencing food insecurity had poorer diet quality and higher likelihood of overweight or obesity, suggesting that constrained food access can coexist with excess body weight due to reliance on cheap, energy-dense foods (Nur Izzati et al., 2022). Beyond Malaysia, a contemporary review of food insecurity in higher education found that food-insecure students reported worse self-rated health and significantly higher levels of depressive symptoms than their food-secure peers, with financial strain and compromised diet identified as key mechanisms linking food insecurity to mental health (Silva et al., 2023). Recent international work also indicates that higher food insecurity scores are associated with greater depressive symptoms and unhealthy lifestyle behaviours, reinforcing the view of food insecurity as a chronic psychosocial stressor that undermines both mental well-being and physical health among young adults (Hernández et al., 2023; Park et al., 2025). The respondents were asked. In this study, respondents were assessed using the 6-item USDA Adult Food Security Survey Module, a widely used short-form measure of adult food insecurity (Sulaiman et al., 2021). This module has been applied in both international and Malaysian research on food insecurity, including studies of university students, indicating that it is acceptable and increasingly common in local settings (Sulaiman et al., 2021; Silva et al., 2023).

2.4.3 Subjective social status

Subjective social status reflects an individual's internal perception of their family's standing in the social hierarchy, a construct that creates key socioeconomic dimensions such as financial security, educational attainment, and occupational prestige. Although objective measures like income and education are fundamental to socioeconomic status (SES), subjective assessments—such as asking respondents to describe their family's financial

situation or rank their social position—often capture the psychological reality of social inequality more proximally than objective indicators alone (Präg et al., 2023; Tan et al., 2024). Research indicates that individuals see these subjective judgments by implicitly aggregating visible markers of status, including parental occupation and the highest level of education in the household, meaning that questions about family wealth, breadwinner education, and job status effectively tap into this overarching construct (Tan et al., 2024; Rahal et al., 2020). Empirically, lower subjective social status is consistently associated with poorer health outcomes, including worse self-rated health and elevated depressive symptoms among adolescents and young adults, largely because the perception of being "less comfortable" or lower-ranked acts as a chronic psychosocial stressor (Rahal et al., 2020; Hoebel et al., 2022). By assessing subjective status through perceptions of family financial comfort alongside breadwinner education and occupation, this study captures the multidimensional nature of social standing, which is hypothesized to be a significant predictor of health outcomes.

2.4.4 Physical exercise

Regular physical exercise is consistently associated with better mental and physical health outcomes, including lower depressive symptoms, healthier BMI, and improved self-rated health. Among college students, exercise is linked to reduced depressive symptoms, with evidence that psychological processes (e.g., rumination) mediate this benefit, suggesting exercise supports mental health both directly and indirectly (Zhu et al., 2024). In Malaysia, national public health guidance emphasizes physical activity as a cornerstone for improving population health and preventing non-communicable diseases, reflecting its role in enhancing perceived health and functional well-being (Khoo et al., 2020). Emerging evidence in adolescents also connects higher physical activity with more favorable BMI profiles and lower depression risk, underscoring exercise's protective role across developmental stages and health domains (Sahril et al., 2025).

While moderate exercise is protective, excessive or poorly managed activity can harm health. A Malaysian study surveying over 500 university students highlighted that high-intensity or unbalanced exercise routines were associated with increased stress and mental health challenges, particularly among those already vulnerable to anxiety and depression. "The negative impact is commonly observed among the elite players as they are required to perform at maximum to get the desired outcomes. As a result, this can cause a negative effect on the mental health of the elitelevel players" (Congsheng et al., 2022). Engaging in physical activity far beyond recommended levels does not necessarily enhance mental well-being and may have adverse effects. Excessive exercise can increase psychological strain and fatigue, potentially offsetting its mental health benefits (da Cunha et al., 2025). Furthermore, extreme exercise loads have been linked to adverse BMI outcomes, including disordered eating and unhealthy weight control behaviors in student populations

(Naidu et al., 2025). These findings underscore that beyond a certain threshold, exercise may correlate negatively with health indicators.

The inverted U-shaped relationship emerges when considering both the positive and negative effects of exercise. At low-to-moderate levels, physical activity improves self-rated health, stabilizes BMI, and reduces depressive symptoms. However, as intensity or frequency increases beyond optimal levels, the benefits plateau and eventually reverse, leading to fatigue, stress, or maladaptive behaviors.

2.4.5 Fruit and vegetables intake

Consumption of fruits and vegetables is significantly associated with BMI, self-rated health and depressive symptoms which are the major behavioural determinants of students' general health. According to an empirical research the infrequent vegetable consumption is linked with worse self-rated health: respondents, who did not take vegetables daily but once a month (OR = 2.19) or weekly, have statistically significant higher odds for reporting poor self-rated health in comparison to those eating it every day showing an evident dose–response association between vegetable intake and perceived health status (Hege PV et al.,2025). Poorer diet quality, i.e., lower intake of health-promoting foods such as fruits and vegetables and more added sugar in the HEI-2020, has also been positively associated with risk indicators for poor health among student populations and females (Zhang et al., 2015), implying a potential preventative approach to reducing both mental/emotional risks as well as physical ones in young adults. It also shows better of functional health, not just in self-perception but also objectively; one weighted logistic regression model found that the middle and high intake groups were significantly less likely to be frail than the low-intake group (OR = 0.70 and 0.68, respectively) which supports fruits and vegetables in physical fitness or robustness for long-term healthy living (Hu et al., 2024). Among university students in Malaysia, healthier dietary habits (i.e., higher fruit and vegetable intake) have been found to be related to better psychological profiles and weight status profiles (Ahmad et al., 2020; Azhar et al. 2023).

In addition to examining health outcomes, this study also addresses a research gap identified in the literature: the factors contributing to food insecurity among students. Accordingly, food insecurity is treated as a dependent variable in a separate model, with several independent variables considered as potential predictors.

2.4.6 Kulliyah

Academic discipline significantly influences student well-being, with distinct correlations found between a student's field of study and their physical and mental health. Research by Lina-Mursyidah et al. (2025) highlights that those in education and human sciences programs are especially vulnerable to elevated depression, stemming from the heavy

intersection of academic rigor, clinical practice, and emotional labor. Physical health disparities are also discipline-dependent; for example, Huaman-Espino et al. (2024) observed that students in Economics and management programs are more prone to obesity and lower healthy lifestyle scores than their counterparts in Health Sciences. According to Lorz and Thies (2025), these trends are often linked to specific "subject-based health trajectories." Students in creative fields, for instance, report more frequent health issues and heightened anxiety regarding their future employability. While some disparities may arise from self-selection into certain majors, the unique demands of a discipline—such as emotional exhaustion or career unpredictability—can act as drivers for food insecurity and psychological stress, ultimately manifesting in poor self-rated health and disordered eating (Lorz & Thies, 2025).

2.4.7 Gender

Gender plays an important role in explaining differences in BMI, self-rated health, and depressive symptoms among university students. Evidence from Malaysian and international studies shows that female students consistently report higher levels of depression and other internalising symptoms, such as insomnia, appetite changes, and emotional distress, compared to males, indicating greater vulnerability to mood-related problems in this group (Lina-Mursyidah et al., 2025). In contrast, national surveys report that women in the general Malaysian population have higher odds of being overweight or obese, yet university-based findings often show male students with higher BMI and less healthy lifestyle behaviours, including lower physical activity and poorer diet quality, suggesting gender-specific patterns in weight-related behaviours and risk (Rajasegaran et al., 2024; Saintila et al., 2024). Gender differences also extend to self-rated health, where female students are more likely to rate their health as good and excellent and to co-report psychological distress, while male students may under-report mental health difficulties despite having higher cardiometabolic risk profiles (Chia et al., 2023; Saintila et al., 2024). Taken together, these findings support the inclusion of gender as a key sociodemographic variable when analysing how food insecurity and related stressors influence BMI, self-rated health, and depressive symptoms in the student population.

2.4.8 Level of study

Level of study is an important factor because students in different years (Level 1–4) face distinct academic demands and stress profiles that can influence BMI, self-rated health, and depressive symptoms. Evidence shows that earlier-year undergraduates often experience high depression, anxiety, and stress during the transition into university and in learning to manage time, workload, and finances, whereas later-year students face greater pressures from final-year projects, internships, and job uncertainty (Mansor et al., 2022; Rosli et al., 2021). At the same time, research on lifestyle behaviour suggests that senior students generally display better self-regulation and health awareness than juniors, reflecting greater maturity in managing their own health behaviours over time (Anu & Muhammad, 2020). Malaysian

studies also indicate that financial strain and food insecurity can increase in middle and senior years as parental support declines and academic costs accumulate, underscoring the need to account for level of study when examining how these stressors relate to health outcomes among undergraduates (Nurul Izzati & Azdie, 2022; Ramli et al., 2023)

2.5 Independent variable (Food insecurity model)

2.5.1 Household income per capita

Household income per capita is a fundamental economic determinant of food insecurity, serving as a direct measure of a family's purchasing power relative to its size. Economic theory and empirical evidence consistently indicate that higher per capita income protects against food insecurity by ensuring sufficient financial resources to purchase adequate, nutritious, and culturally appropriate food (Ahmad et al., 2020; McKay et al., 2025). In the context of Malaysian university students, recent studies reinforce that financial constraints are a primary driver of food insecurity. For instance, Nur Izzati et al. (2022) found that students from families with lower household incomes (e.g., below RM3,000) had a significantly higher prevalence of food insecurity (56.1%) compared to their wealthier peers, highlighting a clear income gradient in food access. Similarly, Faidi et al. (2023) reported that monthly household income and financial aid eligibility were significant predictors of food security status among Malaysian undergraduates, with low income and the high cost of living frequently cited as major barriers to obtaining sufficient food (Faidi et al., 2023). These findings are corroborated by broader population studies in Malaysia, which show that low household income per capita is strongly associated with increased risk of food insecurity and subsequent poor nutritional outcomes, such as stunting and underweight, in low-income communities (Naser et al., 2014; Tay et al., 2022).

2.5.2 Environmental shocks (i.e flood)

Floods exert a multidimensional impact on food security by disrupting availability, access, utilization, and stability, and their effects are often measured through indicators such as frequency of flood occurrences, duration of flood events, and household exposure over time. In Bangladesh, recurrent flooding damages crops, interrupts markets and transportation, and reduces household purchasing power, with women disproportionately affected through higher rates of undernutrition and underweight status (Lotus et al., 2025). Across Africa, quantitative analyses using Granger causality and panel modeling show that approximately 12% of food-insecure populations between 2009 and 2020 had their status directly influenced by floods, with measurement based on flood frequency, affected area, and cumulative duration (Reed et al., 2022). In Malaysia, recurrent major floods have severely disrupted agricultural production by destroying crops, damaging irrigation infrastructure, and reducing overall food supply, as seen in events such as the 2014 Kedah floods and the 2021 Selangor floods. These flood events intensify food insecurity, particularly among low-income agricultural communities, by cutting household incomes and limiting access to affordable food, thereby worsening existing vulnerabilities (Mahmood et al., 2022). Based on the three

articles, we can show that as the frequency or severity of floods increases, the level of food insecurity also increases because it can reduce access to nutritious food, eroding livelihoods, and heightening vulnerability among already at-risk populations.

2.5.3 Extracurricular participation

Participating in extracurricular activities is a vital aspect of university life that can influence food security status through competing demands on time and resources, as well as through the cultivation of social support. While involvement in campus organizations offers opportunities for networking, recent research indicates it can also exacerbate food insecurity by straining students' limited time and financial budgets. For example, international graduate students and undergraduates with heavy non-academic commitments report that balancing these activities with coursework reduces the time available for paid employment or meal preparation, leading to reliance on expensive convenience foods or meal skipping (Henry et al., 2025). In Malaysia, recent studies on university students have similarly identified that financial instability and the high cost of living are compounded by lifestyle factors and time pressures, where students balancing multiple roles are more vulnerable to food insecurity and poor dietary outcomes (Nur Izzati et al., 2022).

Conversely, extracurricular participation acts as a protective factor by building social capital, which facilitates access to food resources through peer networks. Active engagement in student organizations creates support systems where information about food assistance (such as pantries or free meals) is shared more effectively, helping to mitigate severe hunger (Sklar et al., 2025). Furthermore, student-led initiatives and campus-based programs have been shown to reduce stigma and improve food access, with students who are integrated into these social networks being more likely to utilize available resources to maintain their food security (Dharmayani et al., 2025; Soldavini et al., 2022). Thus, extracurricular involvement poses pros and cons as it can induce resource strain but also provides critical social safety nets that buffer against food insecurity.

2.5.4 Courses taken per semester

Academic workload, typically measured by the number of subjects enrolled or credit hours taken per semester, has been consistently linked to students' eating behaviors and their risk of food insecurity. Osiecki et al. (2022) and Celik et al. (2023) show that heavy coursework, tight deadlines, and high academic pressure often push students toward irregular eating patterns, such as skipping meals or relying on inexpensive, convenient foods. These findings, observed among Turkish undergraduates and international student populations, suggest that the structural demands of higher education directly disrupt normal dietary routines. Amin et al. (2023) further strengthens this connection by demonstrating that high academic stress not only undermines meal regularity but also increases psychological strain, reducing students' ability to plan, shop for, and prepare balanced meals. Taken together, these studies highlight that the measurement of workload through subjects or credit hours is not merely an academic indicator but a meaningful predictor of stress-related dietary

compromises. The cumulative evidence suggests that the pressures of academic life, long hours, demanding coursework, and heightened stress, contribute to reduced access to nutritious food options, thereby positioning academic workload as a substantive factor in understanding food insecurity within higher education settings.

2.5.5 Zakat

Zakat in this paper is measured by the amount a student receives per semester. The prevailing literature suggests a negative relationship between Zakat assistance and food insecurity, where increased aid leads to a significant reduction in hunger. Within the Malaysian context, Adnan et al. (2025) and Ahmad et al. (2025) demonstrate that targeted Zakat disbursements for students (Asnaf Fisabilillah) alleviate the "financial shocks" that typically lead to meal skipping. When students receive sufficient funds to cover *daruriyyat* (basic needs), their economic access to nutritious food stabilizes, thereby lowering the prevalence of very low food security status. This is supported by Adnan et al. (2025), who found that Zakat in public universities effectively meets students' survival needs, including food and beverage expenses, allowing them to redirect their focus toward academic performance rather than basic survival.

Conversely, a positive relationship can emerge when the Zakat amount received is insufficient or diverted, actually increasing the likelihood of food insecurity for the student. This occurs because Zakat recipients often belong to the *faqir* or *miskin* categories, where the student serves as a secondary financial pillar for their household. Yahaya et al. (2025) identified that a common pattern among student recipients is "aiding family," where funds intended for the student's personal consumption are instead sent home to alleviate the poverty of their parents in rural areas. This diversion of funds, combined with what Faidi et al. (2023) describe as the "inflation-stipend gap," means that the remaining money is often insufficient to cover the rising cost of on-campus meals. Additionally, Ayub et al. (2023) note that the high cost of digital learning tools and "hidden expenses" of university life often consume Zakat money before it can be used for nutrition, leaving the recipient in a state of "funded but hungry."

The relationship between Zakat assistance and food insecurity in this context is best described as **U-shaped**. At very low levels of Zakat, students remain food insecure because they lack support. As the amount of Zakat increases to a sufficient level, food insecurity decreases since students can cover their basic needs. However, when Zakat is inadequate relative to rising costs or diverted to family obligations, food insecurity rises again. This means food insecurity first falls with aid, then increases once the aid becomes insufficient or misallocated, forming a U-shaped curve.

An **inverted U-shaped relationship** is also possible which means that food insecurity is low when Zakat is very small because students are not yet dependent on aid or they manage through other coping strategies. As Zakat increases to moderate levels, food

insecurity rises because the small stipends create dependency but remain insufficient or are diverted to family obligations. Finally, when Zakat becomes large enough to fully cover basic needs, food insecurity decreases again. In this case, food insecurity is low at both ends and highest in the middle, forming an inverted U-shaped curve.

2.5.6 Scholarship

The three studies clearly highlight a negative relationship between financial scholarships and food insecurity among Malaysian students. Azmi et al. (2022), in their study on Sociodemographic Predictors of Food Insecurity Among Malay University Students, found that sociodemographic and financial background factors such as household income, parental occupation, and number of siblings were significant predictors of food insecurity. Students from lower-income households or with unemployed parents were more likely to experience food insecurity, highlighting that inadequate financial support, including scholarships and family resources, directly worsens food access. Jamil et al. (2020) reinforced this by demonstrating that students with greater financial problems, despite receiving government assistance, were 3.5 times more likely to be food insecure, showing that limited or poorly managed financial support worsens food insecurity. Alias et al. (2024) extended this to urban households, where low income and rising food prices reduced accessibility and quality of food, again underscoring that inadequate financial resources (including scholarships for students) negatively affect food security. In all three cases, the relationship is negative: when scholarships or financial support are insufficient, food insecurity increases. This was measured through self-reported adequacy of scholarships (Razak et al.), structured financial problem and literacy scales (Jamil et al.), and household income categories with food price inflation analysis (Alias et al.). However, we are measuring the scholarships by the amount the respondent receives per semester. Together, they demonstrate that inadequate financial scholarships and assistance directly exacerbate food insecurity among students and low-income households.

3.0 Data and Methods

3.1. Data Collection

IIUM emphasizes the holistic development of students by promoting physical, mental, and spiritual well-being through sports, recreational activities, counseling services, and health awareness programs. As a major Malaysian university with a diverse Muslim student population, including both local and international students, IIUM offers a meaningful context to study health outcomes. Cultural and religious values within this community influence lifestyle, rest, diet, and balance, making IIUM an appropriate setting to explore how these factors shape health behaviors and overall well-being, including both positive and negative aspects. To ensure a relatively balanced number of respondents across all levels, each level was pre-assigned among the researchers prior to the application of the convenience sampling

method, thereby facilitating a more systematic data collection process. Consequently, primary data was collected from a total of 852 respondents to capture the on-the-ground reality of student life. The survey was digitized using Google Forms and disseminated through WhatsApp groups specific to each level of study within KENMS and KIRKHS.

Data for this study were collected using a 30-item questionnaire adapted from Willis (2020), *"Food Inequality: Food Insecurity, Social Status and College Student Health,"* and modified to suit the Malaysian undergraduate context. The questionnaire was administered online via Google Forms and underwent two rounds of consultation to ensure clarity and relevance of the items. A pilot test involving 35 respondents was conducted. Feedback from the pilot study indicated several areas for improvement, including the need to add an "other" option for questions on household occupation to capture a wider range of employment types, concerns regarding the overall length of the questionnaire, and the presence of several confusing items. Based on this feedback, the questionnaire was revised accordingly. followed by a final consultation with the supervisor before full deployment. To achieve a large and balanced sample across two Kulliyyah and levels of study, data collection tasks were divided according to the level of study. Our group was responsible for Level 1 students and successfully collected 221 responses out of a total of 852 respondents (25.9%).

3.2 Methodology

We utilize the Ordinary Least Squares (OLS) regression technique to examine the relationship of Health, BMI, Depressive Symptoms, and Food Insecurity with their respective independent variable In this specification, β_0 is the intercept, $\beta_1,\beta_2,\beta_3\ldots\ \beta_n$ are the coefficients of the independent variables, and ε is the error term.

The OLS regression models are specified as follows:

$$\mathbf{SRHi = \beta_0 - \beta_1 FISi + \beta_2 EDUi + \beta_3 FCi + \beta_4 OCCUi + ln\beta_5 PIi + \beta_6 PEi - \beta_7 PEi^2 + \beta_8 FVi \pm \beta_9 KULi + \beta_{10} GENi + \beta_{11} STUi + \varepsilon}$$

SRHi = Self Rated Health of the *i-th* student, measured using Likert scale ranging from 1 (Poor) to 5 (Excellent)

FISi = Average Food Insecurity Score of the ith student in a week, 1 is never and 5 is always

EDUi = Level of Education of the parents's *i-th* student, measured using an ordinal scale, 1 is high school and 4 is PHd Doctorate

FCi = Financial Comfortability of the *i-th* student, measured using an ordinal scale, 1 is not comfortable and 4 is very comfortable

OCCUi = Occupation of *i-th* student's parents, 1 is not professional and 0 is professional

PIi = Parent's Monthly Income of the *i-th* student, measured as total monthly income in Ringgit Malaysia (RM/month)

PEi = Number of Physical Exercise activities in a week of the *i-th* student

FVi = Average Fruits & Vegetables Intake in a week of the *i-th* student

KULi = Kulliyyah (faculty) of the *i-th* student, 1 for KENMS and 0 otherwise

GENi = Gender of the *i-th* student, 1 for female and 0 otherwise

STUi = Level of Study of the *i-th* student, measured as an ordinal variable reflecting the respondent's year of study

$$\mathbf{BMIi = \beta_0 - \beta_1 FISi + \beta_2 EDUi + \beta_3 FCi + \beta_4 OCCUi + ln\beta_5 PIi + \beta_6 PEi - \beta_7 PEi^2 - \beta_8 FVi \pm \beta_9 KULi + \beta_{10} GENi + \beta_{11} STUi + WORKi + \varepsilon}$$

BMIi = Body Mass Index of the *i-th* students, measured by dividing an individual's body weight in kilograms by the square of their height in meters.
FISi = Average Food Insecurity Score of the ith student, 1 is never and 5 is always

EDUi = Level of Education of the parents's *i-th* student, measured using an ordinal scale, 1 is high school and 4 is PHd Doctorate

FCi = Financial Comfortability of the *i-th* student, measured using an ordinal scale, 1 is not comfortable and 4 is very comfortable

OCCUi = Occupation of *i-th* student's parents, 1 is not professional and 0 is professional

PIi = Parent's Monthly Income of the *i-th* student, measured as total monthly income in Ringgit Malaysia (RM/month)

PEi = Number of Physical Exercise activities in a week of the *i-th* student

FVi = Average Fruits & Vegetables Intake in a week of the *i-th* student

KULi = Kulliyyah (faculty) of the *i-th* student, 1 for KENMS and 0 otherwise

GENi = Gender of the *i-th* student, 1 for female and 0 otherwise

STUi = Level of Study of the *i-th* student, measured as an ordinal variable reflecting the respondent's year of study

WORKi = Number of part-time working hours for *i-th* student per week

$$\mathbf{DSi = \beta_0 + \beta_1 FISi - \beta_2 EDUi - \beta_3 FCi + \beta_4 OCCUi - ln\beta_5 PIi - \beta_6 PEi + \beta_7 PEi^2 - \beta_8 FVi \pm \beta_9 KULi + \beta_{10} GENi + \beta_{11} STUi + \varepsilon}$$

DSi = Depressive symptoms of the *i-th* student, measured by Likert Scale, 1 is never and 5 is always, however, for "I was happy" and "I felt hopeful for the future", 1 for always and 5 is never.

FISi = Average Food Insecurity Score of the ith student, 1 is never and 5 is always

EDUi = Level of Education of the parents's *i-th* student, measured using an ordinal scale, 1 is high school and 4 is PHd Doctorate

FCi = Financial Comfortability of the *i-th* student, measured using an ordinal scale, 1 is not comfortable and 4 is very comfortable

OCCUi = Occupation of *i-th* student's parents, 1 is not professional and 0 is professional

PIi = Parent's Monthly Income of the *i-th* student, measured as total monthly income in Ringgit Malaysia (RM/month)

PEi = Number of Physical Exercise activities in a week of the *i-th* student

FVi = Average Fruits & Vegetables Intake in a week of the *i-th* student

KULi = Kulliyyah (faculty) of the *i-th* student, 1 for KENMS and 0 otherwise

GENi = Gender of the *i-th* student, 1 for female and 0 otherwise

STUi = Level of Study of the *i-th* student, measured as an ordinal variable reflecting the respondent's year of study

$$\mathbf{FISi = \beta_0 - \beta_1 HIPi + \beta_2 FLOODi \pm \beta_3 EXTi + \beta_4 SUBJECTi - \beta_5 ZAKATi + \beta_6 ZAKATi^2 - \beta_7 SCHi + \varepsilon}$$

FISi = Average Food Insecurity Score of the ith student, 1 is never and 5 is always

HIPi = Household income per capita of the *i-th* student, measured by dividing the parents' monthly income (in RM) by the total number of household members.

FLOODi = severity of flood impact experienced by *i-th* student, measured using a 4-point Likert scale, 1 is not at all and 5 is very severe

EXTi = Number of extracurricular activities participated by *i-th* student

SUBJECTi = Number of subject taken by *i-th* student per semester

ZAKATi = Total amount of zakat received by *i-th* student per semester (RM/semester)

SCHi = Total scholarship amount received by *i-th* student per semester (RM/semester).

4.0 Finding and discussion

4.1 Result

$$\mathbf{DSi} = 0.402 + 0.377FISi + \ln 0.100PIi - 0.007EDUi + 0.072STUi -$$
$$(0.336) \quad (0.0247) \quad (0.0399) \quad (0.022) \quad (0.0168)$$
$$0.102FCi - 0.075PEi + 0.015PE^2 i + 0.145\,FVi + 0.019KULi + 0.14GENi -$$
$$(0.037) \quad (0.071) \quad (0.0098) \quad (0.0243) \quad (0.041) \quad (0.042)$$

$n = 852$, $\bar{R}^2 = 0.52$, DW = 1.53, white = 0.00 , p-value = 0.00,

All variable VIF < 5 , except for PEi > 5

DSi = Depressive symptoms of the *i-th* student

FISi = Average Food Insecurity Score of the ith student

PIi = Parent's Monthly Income of the *i-th* student,

EDUi = Level of Education of the parents's *i-th* student

STUi = Level of Study of the *i-th* student

FCi = Financial Comfortability of the *i-th* student

PEi = Number of Physical Exercise activities in a week of the *i-th* student

FVi = Average Fruits & Vegetables Intake in a week of the *i-th* student

KULi = Kulliyyah (faculty) of the *i-th* student

GENi = Gender of the *i-th* student

The regression model yields an Adjusted R-squared ($\bar{R}^2$) value of 0.516, indicating that approximately 51.6% of the variation in depressive symptoms (AVG_CESD_SCORE) is explained by the independent variables after adjusting for the number of predictors. This suggests that the model explains a substantial portion of the variation in depressive symptoms while accounting for the number of predictors included. The F-statistic of 91.96 with a p-value of 0.0000 indicates that the overall model is statistically significant, meaning the independent variables collectively contribute to explaining depressive symptoms. Key variables such as average food insecurity score (FISi), parental income (PIi), and level of study (STUi) show statistically significant coefficients, reflecting the impact of socio-economic factors and food insecurity on depressive symptoms.

The model highlights that average food insecurity score (FISi), parental income (PIi), and level of study (STUi) are statistically significant with p-values less than 0.05, aligning with literature suggesting that socio-economic stressors and food insecurity can influence mental health. However, level of education (EDUi), physical exercise activities (PEi), and kulliyah (KULi) show statistically insignificant effects with p-values above the 0.05 threshold. Specifically: Level of education (EDUi) has a p-value of 0.7386, indicating that it does not have a significant direct effect on depressive symptoms.Physical exercise activities (PEi) has a p-value of 0.2953, indicating that its effect is not statistically significant in this model.Kulliyyah (KULi) has a p-value of 0.6350, suggesting that the faculty of the student does not significantly impact depressive symptoms in this dataset.

The model was tested for potential violations of classical assumptions. The Variance Inflation Factor (VIF) test showed that all variables had VIF values below 5, except for physical exercise activities (PEi), which had a VIF greater than 5, indicating possible multicollinearity with other variables. Despite this, the inclusion of PEi and its squared term (PE^2i) is theoretically justified. The relationship between physical exercise and depressive symptoms is expected to be non-linear, suggesting that the impact of exercise on depressive symptoms could vary at different levels of physical activity. This quadratic relationship means that both PEi and PE^2i should be included to properly capture the diminishing or increasing effect of exercise on depressive symptoms, even though multicollinearity exists. The Breusch-Pagan-Godfrey test and the White test both indicated the presence of heteroskedasticity (p-value = 0.0000), meaning that the variance of the error terms is not constant, which violates the assumption of homoskedasticity. This suggests that the model's coefficient estimates may not be efficient, and standard errors might be biased.

To correct for heteroskedasticity, heteroskedasticity-robust standard errors were applied using the White method. This adjustment corrects for the non-constant variance of the error terms, providing more reliable coefficient estimates and significance levels. While the Breusch-Pagan-Godfrey test and White test still indicated heteroskedasticity, the use of robust standard errors allowed for a more reliable interpretation of the results. Additionally, addressing multicollinearity by noting the high VIF for PEi suggests that further investigation or model refinement may be needed to address potential correlations between physical exercise activities and other socio-economic factors.

Despite the statistical adjustments, variables like physical exercise activities (PEi), level of education (EDUi), and Kulliyyah (KULi) remain statistically insignificant with p-values of 0.2953, 0.7386, and 0.6350, respectively on depressive symptoms in our model may be attributed to various mediating factors. Zhu et al. (2024) found that physical exercise did not significantly predict mental health among college students (B=0.034, P>0.05), even though it significantly reduced other rumination symptoms. This demonstrates that exercise effects are not uniform across all dimensions of depression. Some symptoms respond while others do not, which may explain why your overall depression measure showed no significant relationship. Although educational attainment is generally associated with a lower risk of depression, the relationship becomes statistically insignificant (P>.05) for specific groups such as college graduates compared to high school graduates when the analysis is adjusted to account for adolescent educational aspirations and expectations (Cohen et al., 2020). Finally, the KULi variable's insignificance could be due to contextual factors within faculties, such as

social support and workload stress, which are likely more impactful than faculty name or field of study in determining mental health outcomes (Betancourt-Núñez et al., 2023). Further research with larger or more diverse samples may help clarify these relationships.

4.2 Sensitivity Analysis

To assess the robustness of the baseline regression results, this study conducts a sensitivity analysis by estimating alternative model specifications. These robustness checks are designed to examine whether the main findings remain consistent under different assumptions and variable specifications.

$$\mathbf{SRHi} = 4.16 - \underset{(0.03)}{0.165FISi} + \underset{(0.03)}{0.03EDUi} + \underset{(0.04)}{0.197FCi} - \underset{(0.05)}{\ln 0.035PIi} - \underset{(0.09)}{0.148PE} +$$

$$\underset{(0.01)}{0.03PE^2i} + \underset{(0.03)}{0.05FVi} - \underset{(0.05)}{0.015KULi} - \underset{(0.05)}{0.098GENi} - \underset{(0.02)}{0.108STUi}$$

n= 852 , $\bar{R}^2$ = 12.75, DW = 1.85, white = 0.00, p-value = 0.00

All variable VIF < 5 , except for PEi > 5

SRHi = Self Rated Health of the *i-th* student

FISi = Average Food Insecurity Score of the ith student

EDUi = Level of Education of the parents's *i-th* student

FCi = Financial Comfortability of the *i-th* student

OCCUi = Occupation of *i-th* student's parents

PIi = Parent's Monthly Income of the *i-th* student

PEi = Number of Physical Exercise activities in a week of the *i-th* student

FVi = Average Fruits & Vegetables Intake in a week of the *i-th* student

KULi = Kulliyyah (faculty) of the *i-th* student

GENi = Gender of the *i-th* student

STUi = Level of Study of the *i-th* student

The regression results indicate that the model explains a proportion of variation in self-rated health, with an adjusted R-squared of 0.1275, showing that approximately 12.75% of the variation in self-rated health can be explained after adjusting for the number of explanatory variables. While this value appears low, recent health economics and public health studies emphasize that low adjusted R-squared values are common and acceptable in cross-sectional models involving subjective health outcomes, as self-rated health is influenced by many unobservable psychological, environmental, and genetic factors (Wooldridge, 2020; Johnston et al., 2021, *Journal of Health Economics*). Importantly, the F-statistic of 13.43 with a probability value of 0.0000 indicates that the model is jointly statistically significant, meaning that the explanatory variables collectively have explanatory power even if some individual coefficients are insignificant. (Angrist & Pischke, 2023).

Diagnostic tests reveal a violation of the homoskedasticity assumption, as both the heteroskedasticity test and White test produce p-values of 0.0000, leading to rejection of the null hypothesis of constant error variance. This indicates the presence of heteroskedasticity, which is common in cross-sectional health data where variability in health outcomes differs across income, gender, and work intensity groups (Williams, 2021). Although heteroskedasticity does not bias the OLS coefficient estimates, it renders the conventional standard errors and t-statistics unreliable. In contrast, multicollinearity does not pose a serious concern in this model, as all VIF values are below 5, which is well within the acceptable threshold suggested by recent econometric guidelines. Therefore, the appropriate corrective measure is the use of heteroskedasticity-robust (White) standard errors, which provide consistent inference without altering coefficient estimates.

Some of the variables in the model show statistical insignificance or unexpected signs, which is not unusual and is supported by recent research. For example, the average intake of fruits and vegetables is not statistically significant ($p = 0.0744$), even though we would theoretically expect it to have a positive impact on health. This is likely due to the fact that self-reported dietary information is often prone to errors and recall bias (a systematic error that occurs in research when study participants do not accurately or completely remember past events or experiences), which can weaken the estimated effects in regression models (Subar et al., 2021; Kirkpatrick et al., 2022). Parental education also appears insignificant ($p = 0.3186$). This makes sense because in samples where socioeconomic backgrounds are relatively similar, differences in parents' education may have only a limited direct impact on students' self-rated health. Its influence often operates indirectly, through factors like household income, stress, and lifestyle habits that are already accounted for in the model (Ho & Fenelon, 2022). Similarly, Hageman et al. (2021) found that financial capability (similar to financial comfortability) was not significantly associated with self-reported health ($r = -0.61$, $p < 0.01$ for financial threat, but $p = 0.45$ for financial capability), concluding that while perceived financial threat strongly predicts health, objective financial capability does not. This explains why perceived financial comfort is strongly significant ($p = 0.0002$) while income itself is not.

$$\mathbf{BMIi} = 29.54 - \underset{(0.15)}{0.135FISi} - \underset{(0.29)}{\ln 0.554PIi} + \underset{(0.17)}{0.247EDUi} - \underset{(0.22)}{0.857STUi} + \underset{(0.39)}{0.081FCi} + \underset{(0.81)}{0.11PEi} - \underset{(0.11)}{0.03PEi^2} + \underset{(0.15)}{0.054FVi} - \underset{(0.40)}{0.137KULi} - \underset{(0.43)}{0.893GENi} + \underset{(0.04)}{0.066WORKi}$$

n= 852 , $\bar{R}^2 = 3.549$, DW = 1.94 , white = 0.8505 ,p-value = 0.5835

All variable VIF < 5 , except for PEi > 5

BMIi = Body Mass Index of the *i-th* students

FISi = Average Food Insecurity Score of the ith studen

EDUi = Level of Education of the parents's *i-th* student

FCi = Financial Comfortability of the *i-th* student

OCCUi = Occupation of *i-th* student's parents

PIi = Parent's Monthly Income of the *i-th* student

PEi = Number of Physical Exercise activities in a week of the *i-th* student

FVi = Average Fruits & Vegetables Intake in a week of the *i-th* student

KULi = Kulliyyah (faculty) of the *i-th* student

GENi = Gender of the *i-th* student, 1 for female and 0 otherwise

STUi = Level of Study of the *i-th* student, measured as an ordinal variable reflecting the respondent's year of study

WORKi = Number of part-time working hours for *i-th* student per week

The BMI regression model is jointly statistically significant, as indicated by the F-statistic (F=3.8471,p=0.000020)(F = 3.8471, p = 0.000020)(F=3.8471,p=0.000020), leading to rejection of the null hypothesis that all slope coefficients are jointly equal to zero. This confirms that the independent variables, when considered together, possess statistically significant explanatory power over BMI, even though several individual coefficients are statistically insignificant. Hence, the model is appropriately specified at the joint level.

At the individual level, most lifestyle and socioeconomic variables are statistically insignificant. These include food insecurity scores (p=0.5130)(p = 0.5130)(p=0.5130), physical exercise (PEi) (p=0.8660)(p = 0.8660)(p=0.8660), physical exercise squared (PE^2i) (p=0.7596)(p = 0.7596)(p=0.7596), average fruits and vegetables intake (p=0.7826)(p = 0.7826)(p=0.7826), financial comfortability (p=0.8051)(p = 0.8051)(p=0.8051), level of education (EDUi) (p=0.2353)(p = 0.2353)(p=0.2353), kulliyyah (KULi) (p=0.7261)(p = 0.7261)(p=0.7261), and logged personal income (p=0.1319)(p = 0.1319)(p=0.1319). The insignificance of these variables may be attributed to measurement error in self-reported behaviours, limited variability within the sample, or indirect effects operating through unobserved biological mechanisms such as metabolism or genetic predisposition.

Several variables are statistically insignificant in the BMI regression, which is consistent with previous studies. Average fruit and vegetable intake is insignificant (p=0.0744)(p = 0.0744)(p=0.0744) despite expectations of a negative relationship with BMI. This may be due to measurement error and under-reporting in self-reported dietary data, especially among individuals with higher BMI, which weakens the observed association (Ravelli & Schoeller, 2020).

Similarly, the level of education is insignificant (p=0.3186)(p = 0.3186)(p=0.3186), reflecting evidence that education has a limited direct effect on BMI in relatively homogeneous samples such as students (Witkam et al., 2021). Logged personal income is also insignificant (p=0.4539)(p = 0.4539)(p=0.4539), supporting findings that income often influences BMI indirectly rather than directly through lifestyle and environmental factors (Pourfarzi et al., 2022). In contrast, financial comfortability is significant (p=0.0002)(p = 0.0002)(p=0.0002), suggesting that perceived economic well-being better explains weight-related behaviour among young adults.

Conversely, several variables show significant effects. Level of study is negatively associated with BMI (p=0.0000)(p = 0.0000)(p=0.0000), indicating lower BMI among students at higher

academic levels. Gender is significant (p=0.0259)(p = 0.0259)(p=0.0259), and working hours per week are positively related to BMI (p=0.0281)(p = 0.0281)(p=0.0281), implying that longer working hours may reduce time for healthy behaviours.

Diagnostic tests indicate that the classical OLS assumptions are not violated in this model. The heteroskedasticity tests yield p-values of (p=0.5835)(p = 0.5835)(p=0.5835) and (p=0.5807)(p = 0.5807)(p=0.5807), while the White test reports p-values of (p=0.8505)(p = 0.8505)(p=0.8505) and (p=0.8370)(p = 0.8370)(p=0.8370), all exceeding the 5% significance level. Therefore, the null hypothesis of homoskedasticity cannot be rejected, suggesting constant error variance. Furthermore, multicollinearity is not a concern, as all Variance Inflation Factor (VIF) values are below 5, indicating acceptable correlation levels among the explanatory variables.

Overall, the BMI regression model satisfies the key OLS assumptions, and the estimated coefficients and standard errors can be considered reliable for statistical inference. While BMI appears difficult to explain using socioeconomic and self-reported behavioural variables alone, the significant effects of level of study, gender, and working hours per week provide meaningful insights into the determinants of BMI within the sample. Overall, the sensitivity analysis indicates that the key results of the baseline models are robust across alternative specifications.

4.3 Contribution to the study

$$FISi = 1.203 - 7.76\times10^{-5}HIP\ i + 0.423FLOODIi - 0.030EXTi + 0.045SUBJECTi$$
$$(0.166) \quad (2.5x\ 10^{-05}) \quad (0.037) \quad (0.0136) \quad (0.0242)$$

$$+ 0.003ZAKATi - 1.78\times10^{-6}\,ZAKAT^2i - 5.20\times10^{-0.5}\,SCHi + \varepsilon i$$
$$(0.000322) \quad (1.93x10^{-07}) \quad (3.43x10^{-05})$$

n= 852 , $\bar{R}^2$ = 0.59, DW = 1.57, White Test =0.00 , Breuch Pagan- Godfrey = 0.55

p-value = 0.00, All variable VIF < 5 except for ZAKi VIF > 2

FISi = Average Food Insecurity Score of the ith student in a week ,

HIPi = Household income per capita of the *i-th* student,.

FLOODi = severity of flood impact experienced by *i-th* student,

EXTi = Number of extracurricular activities participated by *i-th* student

SUBJECTi = Number of subject taken by *i-th* student per semester

ZAKATi = Total amount of zakat received by *i-th* student per semester (RM/semester)

SCHi = Total scholarship amount received by *i-th* student per semester (RM/semester).

The regression model provides an Adjusted R-squared ($\bar{R}^2$) value of 0.590, indicating that 59.0% of the variation in food insecurity scores (FISi) is explained by the independent

variables, after adjusting for the number of predictors. This suggests that the model is relatively robust in explaining the variation in food insecurity among students, accounting for the effects of the predictors included. The F-statistic of 178.65 with a p-value of 0.0000 indicates that the overall model is statistically significant, confirming that the independent variables collectively contribute to explaining food insecurity scores. Key variables like income per capita (IPCi), flood impact (FIi), and zakat received (ZAKi) are statistically significant, reflecting the influence of socio-economic and environmental factors on food insecurity.

The model reveals that income per capita (IPCi), flood impact (FIi), and zakat received (ZAKi) are statistically significant with p-values less than 0.05, which aligns with prior research suggesting that socio-economic factors and environmental conditions strongly influence food insecurity. However, subjects taken per semester (SUBi), extracurricular involvement (EXTi), and scholarship per semester (SCHi) all show p-values greater than 0.05, indicating that these variables are statistically insignificant in this model. SUBi has a p-value of 0.0640, while EXTi and SCHi have p-values of 0.0750 and 0.1291, respectively. These results suggest that these academic-related factors do not have a strong direct effect on food insecurity in the context of this dataset.

The model was tested for potential violations of classical assumptions. The Variance Inflation Factor (VIF) test showed no signs of multicollinearity, as all VIF values are below 5, indicating no problematic correlation among the independent variables. However, the White test indicated heteroskedasticity (p-value = 0.0000), meaning that the variance of the residuals is not constant, which violates the assumption of homoskedasticity. This could affect the reliability of coefficient estimates and standard errors. The Breusch-Pagan-Godfrey test did not indicate any issues with heteroskedasticity, showing mixed results. It is important to note that ZAKi, the amount of zakat received by the i-th student, has a VIF value greater than 5, suggesting a potential issue with multicollinearity. Despite this, the theory behind zakat is quadratic in nature. This suggests that both ZAKi and its quadratic form, ZAK2i, should be included in the model to capture the non-linear relationship between zakat and food insecurity. The inclusion of ZAK2i helps to account for diminishing or increasing marginal effects as zakat increases, reflecting the complex impact of zakat on food insecurity. Therefore, while ZAKi shows signs of multicollinearity, its theoretical justification and the need to model the non-linear effect make its inclusion in the model crucial.

To address the heteroskedasticity detected by the White test, heteroskedasticity-robust standard errors were applied using the Huber-White method. This adjustment ensures that the coefficient estimates remain reliable despite the presence of heteroskedasticity. While the White test still indicates heteroskedasticity, the robust standard errors correct for the non-constant variance of the residuals, providing more accurate statistical significance. The

use of these robust standard errors helps to make the model's conclusions more reliable, particularly concerning the significance levels of the variables.

Despite the statistical adjustments, variables like extracurricular involvement (EXTi), subject taken per semester (SUBi), and scholarship per semester (SCHi) remain statistically insignificant with p-values above 0.05. This suggests that their effects on food insecurity may be indirect or mediated by other factors, such as socio-economic status or broader financial stressors. For example, research on university students found that food insecurity was not significantly correlated with extracurricular involvement and similar academic engagements as it may not directly influence food insecurity outcomes (Betancourt-Núñez et al., 2024). Likewise, academic workload emphasizes impacts on mental health and general well-being rather than a direct relationship to food insecurity, suggesting that academic burden alone does not predict food insecurity (Tayoto, 2024). Finally, systematic evidence shows that even with financial aid and scholarship systems available, food insecurity remains widespread among students, justifying that students who receive such aid are still exposed to food insecurity (McKay et al., 2025).

5.0 Limitations

While the econometric equations and E-Views calculations provide a statistically significant foundation, they are limited by their inability to capture the "lived experience" and the complex human behaviors behind the data. Statistical results may not always tell the 100% truth because quantitative models often miss "hidden coping mechanisms". For instance, the food insecurity model might show a student is "food secure" because they have a higher zakat amount, qualitative fieldwork could reveal that the student is practicing "meal stretching"—splitting one plate of rice into two portions to save money for books and other academic materials. Furthermore, interviews can uncover a paradox in academic data; a student might report a high CGPA (3.50 - 4.00) and appear to be thriving, yet in reality, they are suffering from high levels of worry and anxiety about running out of food to maintain that performance. By going to the field and conducting in-depth interviews, future researchers can explain the "why" behind the numbers.

6.0 Conclusion

This study examines the association between food insecurity and student health outcomes through multiple regression models and placing special focus on identifying predictors of food insecurity within a university environment. The findings suggest that all the estimated models are statistically significant except the BMI model, implying food insecurity and its correlated socio-economic factors play a stronger and significant impact on students' subjective health (SRH) and mental health (DS), as compared with body weight condition; thus food insecure is negatively associated with students' self-rated health but positively related to depressive symptoms, while higher subjective social status, parental

income, physical activity level and having more fruit/vegetable intakes relate to healthier dietary outcomes. Crucially, this study contributes to our referenced article by empirically showing that students' food insecurity is not limited to household income, but driven equally by environmental shocks, academic load, participation in social activities, and access to financial support mechanisms like zakat and scholarships revealing the structural and institutional aspects of student vulnerability.

Some useful and feasible suggestions can be given according to the aforementioned findings. At the level of students, student-led initiatives such as peer-supported food sharing projects (for example, KENMS Daily Munch and Iftar Jamaie) and participation in income generating extra-curricular activities (for example BACC car boot sale events and BBAC markets), together with wider mutual aid networks, may help to alleviate immediate food insecurity. In order to further ensure inclusivity and sustainability of such entrepreneurial ventures, every kulliyah could introduce a structured micro–seed capital programme wherein low start-up funds would be awarded to promising student entrepreneurs, helping them commence and grow low-cost business activities within the university enterprise. At the tertiary level, universities could enhance the current initiatives of subsidised cafeterias and association with zakat units to identify and support these vulnerable students. In the university policy framework, the embedding of emergency food or cash assistance systems during periods of environmental stress would directly address some determinants identified in the food insecurity model. Finally, at the policy level, food security could be incorporated into higher education welfare policies at the state or national level in order to provide a more sustained safety net for vulnerable students. Collectively, these suggestions are all interventions that could stop food insecurity and assist those who are food insecure.

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11.0 Appendix

#For 4.1

Dependent Variable: AVG_CESD_SCORE
Method: Least Squares
Date: 01/02/26 Time: 18:44
Sample: 1 852
Included observations: 852
White (HC0) heteroskedasticity-consistent standard errors & covariance

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.401902	0.336676	1.193738	0.2329
FOOD_INSECURITY_SCORES	0.377189	0.024786	15.21769	0.0000
LN_PI	0.099947	0.039987	2.499518	0.0126
LEVEL_OF_EDUCATION	-0.007416	0.022212	-0.333869	0.7386
LEVEL_OF_STUDY	0.072711	0.016816	4.323930	0.0000
FINANCIAL_COMFORTABILITY	-0.102153	0.037482	-2.725376	0.0066
PHYSICAL_EXERCISES	-0.074835	0.071461	-1.047216	0.2953
PHYSICAL_EXERCISES^2	0.015402	0.009839	1.565443	0.1179
AVG_FRUITS_VEGGIES_INTAKE	0.145133	0.024251	5.984521	0.0000
KULLIYYAH	0.019488	0.041033	0.474930	0.6350
GENDER	0.144232	0.042181	3.419388	0.0007

R-squared	0.522329	Mean dependent var	2.502934
Adjusted R-squared	0.516649	S.D. dependent var	0.860111
S.E. of regression	0.597979	Akaike info criterion	1.822305
Sum squared resid	300.7239	Schwarz criterion	1.883600
Log likelihood	-765.3017	Hannan-Quinn criter.	1.845781
F-statistic	91.96259	Durbin-Watson stat	1.534834
Prob(F-statistic)	0.000000	Wald F-statistic	188.6278
Prob(Wald F-statistic)	0.000000		

#For 4.2

Dependent Variable: BMI
Method: Least Squares
Date: 01/02/26 Time: 18:42
Sample: 1 852
Included observations: 852
Huber-White-Hinkley (HC1) heteroskedasticity consistent standard errors and covariance

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	29.53581	2.554142	11.56389	0.0000
FOOD_INSECURITY_SCORES	-0.134543	0.153407	-0.877032	0.3807
PHYSICAL_EXERCISES	0.108805	0.811015	0.134159	0.8933
PHYSICAL_EXERCISES^2	-0.028303	0.109540	-0.258383	0.7962
AVG_FRUITS_VEGGIES_INTAKE	0.054414	0.149958	0.362860	0.7168
FINANCIAL_COMFORTABILITY	0.081467	0.391143	0.208278	0.8351
LEVEL_OF_EDUCATION	0.247397	0.165537	1.494510	0.1354
LEVEL_OF_STUDY	-0.856959	0.221871	-3.862420	0.0001
KULLIYYAH	-0.136896	0.400438	-0.341866	0.7325
GENDER	-0.893342	0.425398	-2.100014	0.0360
LN_PI	-0.553911	0.288907	-1.917263	0.0555
WORKING_HOURS_PER_WEEK	0.066112	0.038221	1.729732	0.0840

R-squared	0.047962	Mean dependent var	22.96936
Adjusted R-squared	0.035495	S.D. dependent var	5.748197
S.E. of regression	5.645258	Akaike info criterion	6.313493
Sum squared resid	26769.91	Schwarz criterion	6.380361
Log likelihood	-2677.548	Hannan-Quinn criter.	6.339104
F-statistic	3.847096	Durbin-Watson stat	1.939208
Prob(F-statistic)	0.000020	Wald F-statistic	6.099461
Prob(Wald F-statistic)	0.000000		

#For 4.3

Dependent Variable: FOOD_INSECURITY_SCORES
Method: Least Squares
Date: 01/02/26 Time: 17:41
Sample: 1 852
Included observations: 852
Huber-White-Hinkley (HC1) heteroskedasticity consistent standard errors and covariance

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	1.202765	0.166070	7.242523	0.0000
INCOME_PER_CAPITA	-7.76E-05	2.50E-05	-3.100577	0.0020
FLOOD_IMPACT	0.423197	0.037035	11.42681	0.0000
EXTRACURRICULAR_INVOLVEMENT_...	-0.024343	0.013656	-1.782552	0.0750
SUBJECT_TAKEN_PER_SEMESTER	0.044939	0.024245	1.853569	0.0641
ZAKAT_PER_SEMESTER	0.003376	0.000322	10.48928	0.0000
ZAKAT_PER_SEMESTER^2	-1.78E-06	1.93E-07	-9.199524	0.0000
SCOLARSHIP_PER_SEMESTER	-5.20E-05	3.43E-05	-1.518972	0.1291

R-squared	0.597052	Mean dependent var	2.226526
Adjusted R-squared	0.593710	S.D. dependent var	1.243739
S.E. of regression	0.792770	Akaike info criterion	2.382778
Sum squared resid	530.4407	Schwarz criterion	2.427356
Log likelihood	-1007.063	Hannan-Quinn criter.	2.399852
F-statistic	178.6520	Durbin-Watson stat	1.569725
Prob(F-statistic)	0.000000	Wald F-statistic	288.2294
Prob(Wald F-statistic)	0.000000		